UCS1001 Critical Thinking and Communicating
Assignment 2a: Problem-Solution Team Proposal

Overview

Type: Teamwork (4 members per team)
Weighting: 30% of Module Grade
Submission Deadline: Week 12 by 11.59 PM, 16 November 2026

Learning outcomes

Through this assignment, you should demonstrate your ability to:

- apply critical thinking and systems thinking to analyse engineering problems;
- conduct and synthesise secondary research from credible sources;
- evaluate alternative solutions using evidence-based reasoning;
- develop feasible and sustainable technical recommendations;
- consider the broader technical, organisational, and stakeholder implications of engineering decisions; and
- communicate technical information effectively through a professional proposal intended for workplace decision-makers

Task Overview

You and your team members are working as junior engineers in an engineering organisation. Your task is to identify a significant technical, operational, or sustainability-related challenge associated with an engineering system, product, process, procedure, or workflow, and propose a feasible and sustainable solution.

Your proposal is intended for submission to a relevant engineering stakeholder (e.g., Engineering Manager, Technical Lead, Department Head, or Project Sponsor) with the aim of persuading management to adopt your team's proposed improvement.

Industry-Oriented Framing

In this assignment, you are writing as an engineering project team proposing a practical technical or process improvement within an organisational context. Your proposal should resemble a professional technical proposal rather than an academic essay. Use clear, concise, and professional language appropriate for engineering managers and technical stakeholders.

Support your proposal with relevant academic, industry, and technical sources. Demonstrate critical thinking by evaluating evidence, considering alternative approaches, and justifying your recommendations. Demonstrate systems thinking by considering how your proposed solution may affect interconnected technical systems, organisational processes, stakeholders, safety, sustainability, and operational performance.

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Critical Inquiry

To strengthen your proposal, evaluate the feasibility and sustainability of your proposed solution through structured team discussions and class activities. Generate and use critical questions to challenge assumptions, evaluate alternative perspectives, identify potential risks, and examine broader system implications. Develop **10-12 critical questions** to guide your inquiry. Integrate the insights gained throughout your proposal to demonstrate your critical and systems thinking and include **the 10-12 critical questions** as an **appendix**.

From Ideas to a Proposal

Once your team has identified a suitable engineering challenge, conduct further research using credible academic, technical, and industry sources to understand the current state of the problem and existing solutions.

Evaluate potential solutions based on their technical feasibility, practicality, safety, sustainability, and organisational impact. Consider how the proposed solution may influence engineering performance, operational efficiency, maintenance, users, and other interconnected systems. These considerations should be reflected throughout your proposal.

Suitable topics may include improvements to engineering systems, automation, embedded systems, energy efficiency, electrical or electronic design, manufacturing processes, maintenance practices, workplace safety, cybersecurity, data management, sustainability initiatives, or other engineering-related challenges relevant to your discipline.

Structure of the Team Proposal

Your Team Proposal should include the following sections (with sub-sections where appropriate):

- Executive Summary
- Table of Contents
- Introduction
  - Background
  - Problem Statement
  - Purpose
  - Scope
- Identified Problem and Proposed Solution
- Analysis and Evaluation
  - Technical Feasibility
  - Benefits
  - Limitations
  - Implementation Considerations
- Conclusion
- References
- Appendices

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Length: The main body of the proposal must be approximately **1800 to 2000 words** (from the Introduction through the Conclusion).

Tone & Style: The writing must maintain a professional and formal tone.

Teamwork and Submission Instructions

1. Distribute the workload and collaborate effectively. Use the suggested Proposal Template to ensure all required components are covered.
2. All information and research used must be formally acknowledged to ensure ethical sourcing and academic integrity.
   - **Approved Citation Formats:** Use the 7th Edition APA Style or the IEEE format consistently for all in-text citations and the final References list, as approved by your lecturer. Consult your lecturer immediately to confirm which one (APA 7th Edition OR IEEE) is the definitive format for your submission.
3. **Ethical Use of Sources:**
   - **Paraphrasing and Summarizing:** Ensure all content is properly paraphrased and summarized in your own words.
   - **Direct Quotes:** Use direct quotes sparingly. When used, enclose them in quotation marks and include the specific page or paragraph number.
   - **AI-Generated Content:** If you use generative AI tools (e.g., ChatGPT) for brainstorming or background research, you must treat the output as a source. Verify any facts provided by the AI before including them and include a proper citation for the AI tool in your References list (following the APA or IEEE guidelines for software/tools).

4. **Assessment Criteria:**

- Clarity, organization, and coherence
- Application of critical thinking and systems thinking frameworks
- Persuasiveness and reader usability
- Research depth and citation accuracy
- Language, grammar, and technical accuracy
- Citations & ethical use of information.

Note: Refer to the rubric for a detailed explanation of the criteria.

Collaboration and Accountability: Your contribution to the team's success is a mandatory component of class participation (student engagement) grade. This is assessed based on whether you consistently fulfil your assigned responsibilities, actively contribute ideas and resources to the group effort, and constructively engage with the feedback process.

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Submission Details:

- **Deadline:** By 11.30 PM, 15 November 2026
- **One submission per Team**
- **Format:** Word document (12pt Times New Roman/Calibri/Arial, double-spaced)
- **Submission Platform:** DropBox- XSITe

Attach the Assignment Cover Page and GenAI declaration form to your final draft and submit the softcopy to XSITe

Please note your assignment will automatically be checked by anti-plagiarism software such as Turnitin. Cases of plagiarism and collusion will be penalised heavily. Refer to the SIT handbook on Academic Integrity for more details.

Late submissions for any assignment, without penalties, require a written application to be submitted prior to the submission deadline. If the right for a late submission is not granted, the late submission will then be accepted until 4 days after the original deadline, with a penalty of 15% per day. No marks will be awarded to submissions 4 days after the original deadline.

Critical & Systems Thinking Frameworks:

Paul-Elder Critical Thinking Framework — University of Louisville Ideas To Action

How do you use systems thinking and engineering in your system design?

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